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Optimisation of freight flows and sourcing in sustainable production and transportation networks

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ABSTRACT

Global supply chains require integrated cost optimisation of sourcing, production and distribution through flexible networks of suppliers, manufacturers and logistics operators. The minimisation of environmental impacts of freight flows must be considered for sustainable development and should turn into a competitive advantage. To these aims, the paper presents a model integrating supply, production networks and sustainable freight transportation for strategic and tactical decision-making. Bill-of-Materials constraints are included in the model. The objective function considers sourcing, production and transportation costs as well as carbon dioxide emissions as environmental impacts of transport over a multimodal network. Computational experiments demonstrate the effectiveness of the model for cost-emissions analysis.

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1. Introduction

Sustainability is becoming the pillar of business models based on technological, social and organisational innovations grouping business model archetypes such as, e.g., maximisation of material and energy efficiency, delivery of functionality rather than product ownership, firm responsibility (Bocken et al., 2014). In the medium-long term, the reduction in energy, fuel and material consumption leading to reduced CO_2 emissions can turn into remarkable cost advantages due to direct savings in operations and lower impacts on companies of carbon footprint regulations. At the same time, enhanced value proposition and delivery to customers can be achieved through transparent environmental and social benefits acknowledged and rewarded by stakeholders and markets.

Sustainability is then the key driver for designing and managing global production networks. Current challenges for networked production are to stay competitive by optimising system-wide costs and service levels as well as to eliminate or minimise negative impacts of freight transport on environment and society. These objectives are often separately pursued or only partially integrated in research approaches as well as in planning tools used by industry. It becomes imperative to completely include and widely spread in the industrial practice effective methods and tools dedicated to the accomplishment of business objectives based on sustainability-driven strategies and operations.

Generally, strategic decisions concern the network configuration in terms of plant location, material and component supply selection. Tactical decisions involve lead time considerations and allocation of production and supply quantities to the network; moreover modal transport selection is included at this decision level.

To these aims, the article presents a model integrating production networks and sustainable freight transportation for strategic and tactical decisions. The optimisation model includes Bill-of-Materials (BOM) constraints with alternative supply sources having different production and distribution costs. The objective function concurrently considers sourcing, production and transport operating costs as well as the environmental impact of transport over a multimodal network. Multimodality is represented through a specific graph model. Supply chain lead time constraints for sourcing, production and transportation management are considered. Computational experiments are based on a multimodal network that considers road, rail and sea transport modes. The integrated effect of sustainable freight transport and efficient production network configuration is discussed.

The remainder of the article is organised as follows. Section 2 presents the state of the art of optimisation models integrating production and transportation problems. Section 3 discusses the integration of sustainability metrics and the work contribution.

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In Section 4, problem definition and methodological approach are included while describing the optimisation model. Section 5 presents the test instances while Section 6 illustrates the computational results. In the subsequent sections, the findings are discussed and conclusions follow together with future research tasks.

2. State-of-the-art

Supply chain management (SCM) coordinates strategies, relationships and activities integrating multiple companies across networks consisting of suppliers, manufacturers, distributors, retailers and service providers to achieve efficiency, effectiveness, service level targets, and deliver value to customers. Within the wider scope of SCM, logistics management (LM) plans and controls material and information flows within and between facilities to realise SCM performance goals. The following state-of-the-art literature review is presented following two basic streams that regard the integration of production (including sourcing) and transportation networks:

(1) the integration of production and transportation networks considering production and logistics-related key performance indicators and constraints in the developed models and
(2) the inclusion of environmental sustainability metrics in the integrated production–distribution models.

2.1. Integrated production and transportation

In the operations research and management science literature, the study of integrated production and transportation networks mainly addressed the optimisation of supply, production, inventory, facility, and transportation costs so far. For these purposes, formulations based on multi-commodity production–distribution (Geoffrion and Graves, 1974) have been extensively used. Several reviews and classifications have been proposed in the research literature. More specifically, a classification of models including their solution techniques for production–distribution planning has been recently presented (Fahimnia et al., 2013). The proposed classification hierarchically considers the degree of complexity of the production–distribution models by introducing seven categories which are based on the combination of single or multiple products, single and multiple manufacturing plants, no or multiple warehouses, single or multiple transport paths, no, single or multiple end users, no or multiple time periods. The classification of models based on the solution approaches then includes: (i) mathematical techniques (linear programming models, mixed integer programming models, and Lagrangian relaxation models), (ii) heuristic techniques, (iii) simulation modelling, and (iv) genetic algorithms (Fahimnia et al., 2013). Martel (2005) overviews the design process of production–distribution networks with reference to capital investment planning for the capacity extension, technology selection, facility location and resource allocation in manufacturing and distribution. A mathematical programming approach for designing integrated distribution and manufacturing networks based on make-to-stock models is then proposed. Goetschalckx et al. (2002) review mathematical programming models for global logistics systems while highlighting international characteristics of strategic models published at that time. The research is related to the saving potentials deriving from the integration of the global supply chain design with tactical production–distribution allocations and transfer prices. The integration of production and distribution problems in global supply chains has been reviewed also by Bilgen and Ozkarahan (2004). Yan et al. (2003) propose a strategic production–distribution model for the design of supply chain by using a mixed integer linear programming (MILP)-based model. The included BOM in the formulation makes it possible to explicitly consider the supplier selection in the design process of the supply chain.

Until 2004, only a few research works included freight flow constraints for the BOM in the optimisation models (Bilgen and Ozkarahan, 2004). Finally, Eskigun et al. (2005) propose a capacitated network design model for the outbound distribution of a large-company case study in the automotive industry by using an integer linear programming and a Lagrangian solution approach. Lead times to include customer satisfaction in the analysis, location of distribution facilities and choice of transportation mode are simultaneously considered.

In order to fully integrate production and transportation, a number of formulations for freight transportation planning using different transportation modes are already available, albeit further developments are still needed. SteadieSeifi et al. (2014) present a recent literature review of models for multimodal freight transportation planning by using a decision level-based classification. Concerning their review, it is relevant to point out: (i) at strategic level, the different options for origin-destination transportation (e.g., hub-and-spoke); and (ii) at tactical level the existence of two groups of models for, respectively, network flow planning problems and service network design problems.

2.2. Consideration of environmental sustainability in integrated production and transportation

Sustainability in supply chains is addressed in general terms in the well-known research stream of sustainable SCM which concerns the economic, environmental and social dimensions (Seuring and Müller, 2008). With particular reference to the environmental aspects of product design, sourcing, manufacturing processes, distribution to consumers, product end-of-life management, the research stream of green SCM has been developed (Srivastava, 2007). Sustainable and green SCM encompass the wider dimension of cooperation and performance alignment among supply-chain partners pointing to shared sustainability goals as well as the inclusion of environmentally friendlier purchasing, manufacturing, R&D and distribution (Ageron et al., 2012). At LM level, the realisation of these concepts then requires embedding in methods and tools for planning and control material and information flows concrete environmental metrics to be optimised.

Environmental sustainability has been considered through quantitative approaches in several research works. The literature however seems to be still limited although a promising growth can be observed. Le and Lee (2013) investigate a multi-product supply chain design problem built around a focal company through an integer linear programming model which minimises the cost associated to inventory, shipment quantity, transportation routes and modes as well as CO_2 emissions. The results highlight the influence of CO_2 emissions on decisions regarding inventories, travel times and mode selection. The model is strategic and does not consider the BOM. Dotoli et al. (2006) study an integrated e-supply chain through integer linear programming for network design and optimisation purposes. Two problems are tackled: the multi-objective optimisation of (i) costs, energy, CO_2 emission and total lead time, and (ii) cost, energy and CO_2 emissions. BOM is included in the constraints. While only an aggregated lead time is considered (i.e., the time elapsed from the shipment of raw material to the final delivery of finished goods), other aspects such as the transportation connections are not considered. A supply chain network design problem is tackled by Wang et al. (2011) who propose a multi-objective, mixed-integer formulation that includes within two objective functions, respectively, an environmental protection variable for investment decisions and CO_2 emissions. BOM, lead times and different transportation modes are not considered. In Sundarakani et al. (2010), carbon footprints are modelled while studying a

closed-loop end-to-end supply chain through a transport modelling approach. The model does not include production or logistics-oriented performance metrics or constraints of the supply chain. Stecca et al. (2013), on the basis of previous preliminary models, develop a MILP model of a multimodal network that integrates, respectively, production and logistics plants while including BOM constraints and CO_2 emissions for each transportation mean and sourcing selection. Chaabane et al. (2011) formulate and solve a MILP model for supply chain design that includes logistic costs (fixed, variable and carbon credits) as well as the emission of greenhouse gases in terms of tons of CO_2 equivalent. The integrated model considers transportation modes, suppliers and the BOM. The model has a strategic character and does not tackle factors such as, e.g., the integration of different transportation modes or production models. Shaw et al. (2013) formulate a multi-objective goal programming model for the optimal design of a single product, multi-period supply chain network. Total costs, trade credits as well as carbon emissions in terms of embodied carbon footprints of raw material, manufacturing and transportation along the supply chain are considered. BOM and multimodal transport are not considered. Other research works make use of multi-objective programming and data envelopment analysis for designing efficient, sustainable logistics networks from the standpoints of costs and environmental impact (Quariguasi Frota Neto et al., 2008).

2.3. Insights from the literature review

Concerning the optimisation of production–distribution planning, future research directions have been recently suggested by Fahimnia et al. (2013). On one hand, some of them refer to realistic modelling of scenarios and effectiveness of the related solution techniques, ownership issues as well as volatility and uncertainty of supply chains. On the other hand, the consideration and development of multiple performance metrics addressing in the objective functions, e.g., cost, service level, social and environmental impacts, require further research. Moreover, the inclusion in model formulations of multiple transport paths and shipment modes has been recognised as a research trend in the last five years, although multiple modes only for addressing alternative shipment costs and economies of scale (Fahimnia et al., 2013).

Furthermore, according to the most recent literature, the inclusion of external suppliers and transportation modes in the design of global production and distribution networks, among various aspects, is still very limited and needs to be enhanced in future research (Olhager et al., 2013). In fact, supply (sub-)networks for the acquisition of raw materials, parts, etc. are seldom considered. Additionally, it is important to point out that in the field of transportation planning promising formulations have been developed.

Concerning the latter point, a recent review by SteadieSeifi et al. (2014) on freight transportation planning provides very interesting insights for addressing future research issues that can foster the use of multimodal transport and therefore the sustainability of modern supply chain networks. Important findings of their study, among many others, are that

- For strategic planning problems, future research opportunities should address capacity limitations on links and hubs (that can cause congestion effects), and transhipment with the related costs.
- For tactical planning problems, asset management in transportation service network design, transhipment of loads and related performance impacts, uncertainty and disruptions should be strengthened in future research.

As it can be noticed on the basis of the research literature, only in a few research works the freight flow constraints related to the BOM in the MILP formulations have been used, as already pointed out by Bilgen and Ozkarahan (2004). In these networked production scenarios little evidence has been identified about the sourcing options from alternative suppliers except for a few, very recent works. In the MILP-based models that addressed environmental sustainability, BOM constraints, sourcing options as well as transportation planning are rarely included in an integrated fashion and almost never simultaneously, with the exception of very few recent attempts. It then seems that research lines faced so far the design and management of freight flows across integrated production and logistics networks while focusing on and emphasising specific perspectives related to the production, transport or environmental performance, or only partially considering the combined optimisation of these objectives. Moreover, the research literature is predominantly focused on the analysis of integrated production and transportation from a strategic viewpoint.

3. Sustainability metrics and work contribution

Generally, the integration of sustainability metrics of supply chains in performance management systems still represents a hurdle due to (i) immature research, (ii) scarce evidence of industrial applications (except for sustainability reports of single firms), (iii) difficult, concurrent consideration and single unit of measurement of environmental, social and economic dimensions (Taticchi et al., 2013). Additional hurdles such as, e.g., indicator selection, coordination, different supply-chain players, etc., are emphasised by Hassini et al. (2012). Concerning in particular CO_2 emissions as green key performance indicator (KPI) used in the present work, there could be significant blockers in the usage of such a KPI. Transport mode selection (and therefore the emissions in delivery operations) is influenced by time and cost constraints of decision makers (Hassini et al., 2012). Although rail transport is environmentally friendlier than road, air and some specific waterborne transport, it is not economically viable for very short distances, usually slower and consequently not suitable for the required time reliability of deliveries in supply chains. The same lead time consideration applies to waterway/maritime transport while exposing supply chains to higher risks in terms of capacity to timely react in the event of demand changes or supply chain disruptions, and potentially higher inventory levels together with higher emission levels due to larger production volumes and item obsolescence. Furthermore, supply chain partners such as globally distributed suppliers and logistics service providers (who in turn often rely on outsourcing) should share the same data and green KPI measurement system and targets, which raise additional concerns regarding coordination and reliability of measurement. Nonetheless, there are relevant enablers for the adoption of a CO_2 emission green KPI:

- Availability of CO_2 emission data depending on energy sources.
- Availability of consumption data and meters for each production site and logistics facility.
- Availability of very detailed CO_2 emission data of freight transport (see, for example, French Republic – Ministry of Ecology, Sustainable Development and Energy, and Environment and Energy Management Agency (2012)).
- Enterprise-wide information systems such as ERP as well as track and trace systems that support supply-chain collaboration, data exchange and transport monitoring.
- Regulations which may oblige companies to adopt such measurement systems but may also turn into an opportunity to gain competitive advantage because of enhancements in energy savings and firm reputation.
- Standards such as, e.g., ISO 14,000 series, which may incentivise supply-chain partners to implement CO_2 emission measurement especially if requested by focal companies.

- Third-party logistics providers specialized in particular industries (e.g., automotive) who are able to plan, coordinate, monitor and control multiple transport operations and large fleets on short, medium and long range distances.

This work contributes to the enhancement of decision-making processes in sustainable supply chains by integrating in the objective function a green KPI, namely CO_2 emissions, in the selection of supply sources, assembly plants as well as transport modes for optimally moving components and finished products. It builds upon preliminary models (Galiano et al., 2010) which consider merging of short-haul and long-haul freight transport. Further developments introduced by Stecca et al. (2013) and Liotta et al. (in press) deal with BOM modelling in the formulation considering also suppliers and introduce a sustainability CO_2 factor in a transportation optimisation model. The present work extends the model developed in Stecca et al. (2013) by significantly improving the formulation in terms of objective function (CO_2 emission, costs) and constraints (BOM, time, capacity, modal transport and shift) and creating a structured instance inspired by publicly available industry data in order to test the model. The model mainly deals with planning aspects typical of LM while including different multimodal transport options. Besides, it addresses SCM aspects because of the strategic decisions related to selection and location of suppliers, implicit coordination with transportation operators, as well as allocation of production to markets.

This model aims to be worthwhile for researchers, practitioners and policy makers willing to broaden further the analysis of supply chain configurations driven by multiple sustainability considerations. More specifically, it can bring added-value to researchers engaged in (i) investigating future developments of global and local manufacturing together with new performance management systems for sustainability, and (ii) improving model formulations for sustainability-driven facility location problem or transport mode selection over multimodal networks for multiple items. It aims to benefit practitioners engaged in enhancing location and transport selections as well as intelligent software tools currently used to support these strategic and tactical decisions through green KPIs that can be monitored and controlled. Finally, this work can help policy makers decide measures and investments to (i) to foster the sustainable growth of local/regional territories, and (ii) incentivise and streamline the usage of transport modes with lower impacts on environment and society (e.g., through regulatory frameworks and investments in automated solutions for fast modal shift, increased speed and capillarity of delivery, carbon footprint taxation).

4. Problem and methodological approach

4.1. Problem definition

On the basis of the insights from the existing literature, the main issues to be addressed in our problem formulation consist of

- The simultaneous consideration in the objective functions of factors representing economic and environmental dimensions of sustainability.
- From the standpoint of the decision support level, the simultaneous consideration of (i) strategic decisions, such as the supplier selection in terms of sourcing production plants and the decision on which market has to be served by specific plants, and (ii) tactical aspects, such as the supply, production and transportation lead times as well as the allocation of the quantities to be produced to single suppliers. The available production capacity for each supplier and manufacturing plant is also considered.

Therefore, given the production capacity of production plants, the sourcing, production and transportation cost parameters, the environmental impact factor, and the quantities to be produced for the on-time order fulfilment under the assumption of zero-inventory, the problem is to determine the combination of production sites and multimodal routes over an integrated production and transportation network that concurrently minimises the total production and supply costs according to the BOM requirements, the transportation costs as well as the CO_2 emissions of transportation means used to move materials, semi-finished and finished products. Zero-inventory strategy means that demand requests are fully propagated upstream with no inventory intermediation. With respect to the literature review, our approach considers both economic and environmental aspects of the problem; moreover it considers both strategic (e.g., selection of suppliers and production sites) and tactical aspects (e.g. selection of carriers, lead time, and quantities allocated to each supplier and producer).

4.2. Modelling the production network

The proposed approach considers a production network model in which components (subassemblies) are produced and assembled in different plants in order to build the required final products. In the production and transportation network depicted in Fig. 1, the supply and assembly nodes represent production plants linked by different transport modes. The final nodes correspond to the market that is divided into demand areas. Generally, supply nodes can provide the assembly plants only with a subset of the required components. Each assembly plant can potentially produce all the finished products to be delivered to the demand areas. In Fig. 1, the material flows between supply, assembly plants and demand areas are represented by arcs. Each component or product flow is characterised by a specific dashed line. Intermediate logistics nodes between plants and demand areas are also considered in the optimisation model. The intermediate logistics nodes can be intermodal terminals or other logistics platforms for warehousing and shipment. In the model, multimodal transportation includes three modal options, i.e., road, rail and short-sea shipping modes. When multimodality is considered, virtual nodes and arcs are added to the network in order to represent the modal change. The model considers the demand from the market in order to trigger the assembly and supply operations. For instance, when the customers in demand area 1 issue an order for a final product, a shipment for the final product is associated from the assembly plant 1, 2 or 3 to the node 1. Under the hypothesis of a make-to-order or source-to-order production model and zero-inventory, upon each request of final product k from an origin o_k (assembly plant 1, 2 or 3 in Fig. 1) to a destination d_k, (i.e., demand areas), correspondent orders for the components to be produced or assembled upstream in the supply and assembly plants are issued. Fig. 1 represents a hypothetical supply chain network scenario of the automotive industry. Supply options include the set of available suppliers for each component while assembly options consist of available assembly plants to serve the demand areas.

4.3. Production network model with consideration of BOM and supply network

In order to consider the assembly relation among the different component or product flows in the network, a representation of the BOM must be associated to the production and logistics network. The production/assembly nodes in Fig. 1 are put into a

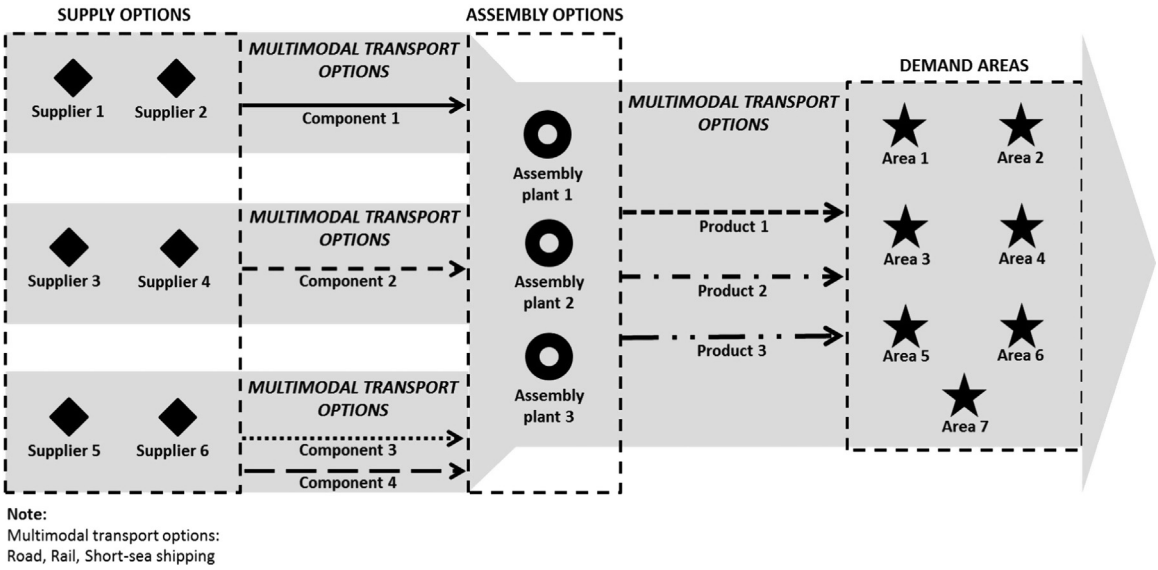


Fig. 1. Production and logistics network including products and alternative sources for each component linked by multimodal transport.

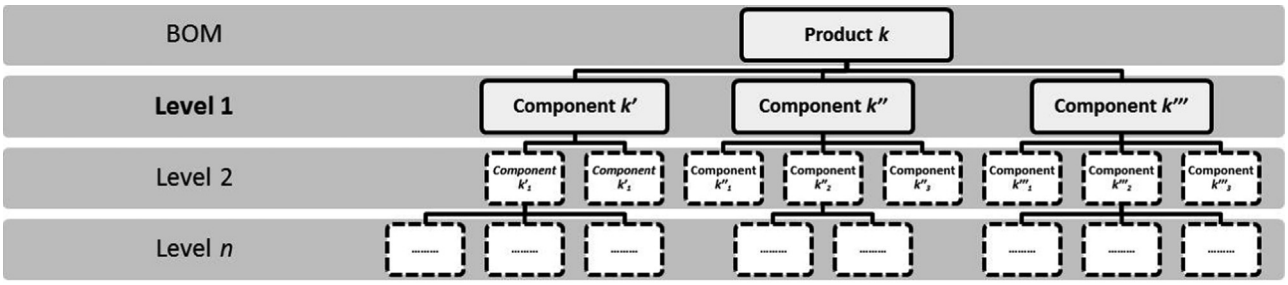


Fig. 2. Product BOMs considered for the scenario tests (Products 1, 2, and 3).

"BOM relation" and must be considered in the mathematical model as flow and time constraints. The representation has to consider also alternative sources for the same component in a specific position of the BOM. The BOMs considered in scenario tests are presented in Fig. 2. A specific BOM corresponds to a specific product. The first level of three different BOMs is considered in this work as highlighted in Fig. 2 (see also Section 4).

The "BOM relation" relies on the fact that each component or product k requested (primarily) by the market, and in turn by the assembly plants, is linked through origin and destination locations to the corresponding BOM. Accordingly, the potential sites where the specific operations needed for each product component included in the BOM are activated. The available transportation options among the various sites allow the material flow over the network, from suppliers to assembly plants and demand areas. Fig. 1 integrates the production and logistics network with the BOMs presented in Fig. 2. Only specific supply sites are available for the production of specific components while the assembly plants can potentially realise all the products and serve the different demand areas.

4.4. Integrated production and transportation networks

Following the general approaches known in the scientific literature, multimodality for shipments is considered at graph level. This means, that transportation modes are drawn by adding proper arcs to the graph model as done in Moccia et al. (2011) and Galiano et al. (2010). Each physical arc is duplicated to represent a single mode of transport. Modal shifts are modelled by duplicating nodes and drawing virtual arcs representing the mode change. In this way an arc (i,j) of the graph represents a connection between two nodes offered by a specific freight transport operator using a defined transportation mode. This choice allows separating the costing and the sustainability considerations for each transportation mode.

Sustainability factors in logistics and manufacturing modelling can be considered under different dimensions. In the proposed model, which extends the work by Stecca et al. (2013), sustainability factors are considered at objective function level. The objective function will minimise transportations costs and CO_2 emitted by the used vehicles, depending on the selected transport mode. A sustainability factor comes from the CO_2 emissions of each transportation vehicle. The emission factor e_{ij} is associated to each arc $(i,j) \in A$ of the network (i.e., to each transportation mode).

4.5. Optimisation model and sustainability factors

Generally, in the scientific literature, multimodality is modelled as an aspect of the classical multi-commodity flow problem. Gendron et al. (1998) model a multimodal transportation problem as a multi-commodity capacitated network design problem defined on a directed graph $G=(N, A)$. In the modelled problem, a set of commodities (shipments) K have to be routed according to known demands and a set of facilities have to be installed on arcs (arc-based formulation). This model was used in different variants in

order to solve transportation problems over integrated networks for short-haul and long-haul freight transport (see, for instance, Moccia et al. (2011) and Galiano et al. (2010)). In the proposed approach, arcs and nodes are replicated in order to have a single arc for each multimodal transportation mean (e.g., a specific scheduled rail service). With respect to the mentioned formulations, K commodities are originated from specific, different sources according to the hierarchical product decomposition included in the BOM as also described in Stecca et al. (2013) and Liotta et al. (in press). BOM constraints are exploited to model the relations among the different layers of the distributed production system allowing two major modelling features:

- connect each layer and node of the product BOM, to the available set of suppliers or sub-assembly manufacturer (thus allowing source selection) and
- synchronise the different layers of the production and distribution network through the BOM time constraints allowing control through lead time constraints at strategic and tactical level.

A further feature included in the model is the representation of different transportation means which can be used in order to allow the selection among a set of carrier options or transportation services. A transportation service is a capacitated, single modal transportation operation that can be carried out within a specific time window by a transport operator. To each transport service an arc is associated between a pair of nodes. Arc and nodes are replicated for each available transportation service. The optimisation procedure simultaneously minimises supply and assembly costs as well as the transportation costs including the contribution of the environmental impact.

For each node $i \in N$ let $\Delta^+(i)$ the nodes $j \in N$ such that $(i, j) \in A$ and $\Delta^-(i)$, the nodes $j \in N$ such that $(j, i) \in A$. $\overline{K} \subseteq K$ be the shipments of the final products. The lead time of the supply chain is the time spent to cross the network from the selected suppliers to the final destinations. The flow is time constrained to be lower than a given maximum lead time L and greater than a minimum lead time E. These bounds are assumed to be connected to specific requirements (e.g., financial considerations, contracts, delivery requirements) imposing limits in the time spent in the supply chain for products and components. Time constraints are also defined for nodes representing production plants. Each operating production plant is constrained to be active for a time falling in an allocated interval $[a_i, b_i]$. The active time of a production plant is computed multiplying the quantity to be produced by the cycle time. These constraints somehow overlap with capacity constrains but are necessary because cycle time may be different for each type of production k and for each plant i; cycle time is in fact defined as the parameter r_{ik} for each node i and shipment k. Let moreover be s_i the service time at node i, for each $i \in N$; t_{ijk} the time needed to traverse an arc $(i, j) \in A$ for shipment k; lts_k is expression of inventory in time units. lts_k is only used for the purpose of time synchronisation of shipments. Each order from demand areas triggers corresponding orders of products and components upstream in the supply chain while no inventory variables are used to satisfy the demand. Other parameters of the model are: M, a large constant; C_{ij} the capacity of arcs $(i,j) \in A$; PC_{ik} the production capacity of node i for product k c_{ij} the cost of traversing arc $(i, j) \in A$; f_i the activation cost of production node $i \in O$; fr_{ijk} fixed routing cost through arc (i, j) for shipment k; p_{ik} the production cost of node $i \in N$ per unit of product k; wsu_k the weight of a unit of shipments of k in tons; α the cost of a ton of CO_2 emission. Emission factors are represented by the parameter e_{ij} where $(i,j) \in A$



Fig. 3. Geographic location of suppliers, assemblers, and demand areas.

Table 1
Demand for products and components.

Demand for products andcomponents							
(Units/year)	Prod. 1	Prod. 2	Prod. 3	Comp. 1	Comp. 2	Comp. 3	Comp. 4
k	*1*	*2*	*3*	*4*	*5*	*6*	*7*
Dem. area 1	25,000	15,000	20,000	60,000	40,000	45,000	35,000
Dem. area 2	35,000	20,000	25,000	80,000	55,000	60,000	45,000
Dem. area 3	15,000	15,000	50,000	80,000	30,000	65,000	65,000
Dem. area 4	50,000	10,000	15,000	75,000	60,000	65,000	25,000
Dem. area 5	40,000	25,000	10,000	75,000	65,000	50,000	35,000
Dem. area 6	45,000	30,000	20,000	95,000	75,000	65,000	50,000
Dem. area 7	65,000	20,000	25,000	110,000	85,000	90,000	45,000
Total	**275,000**	**135,000**	**165,000**	**575,000**	**410,000**	**440,000**	**300,000**

Prod.=product, Dem.=demand and Comp.=component.

and $k \in K$; the dimension of e_{ij} is tonnes of CO_2 for unit of shipment. In the proposed model, the k shipments are linked due to the BOM structural constraints. To this aim we define a set $B \subset K \times K$ containing the precedence constraints between shipments which define the BOM structure. A tuple $(h,k) \in B$ defines the shipment h as a requirement (a material or a subassembly) for the shipment k (a final product or a subassembly).

The model considers alternative sources (i.e., suppliers) for materials and assemblies. To each shipment $k \in K$ is associated a subset of alternative origin nodes $O^k \subset N$ in which the shipment can be produced with $O = \cup O^k$. D^k, $k \in K$ is the set of destination nodes for shipment k, while q_d is the quantity of shipment k to be shipped in destination $d \in D^k$; I^k are the subsets of intermediate nodes for shipment k with $I = \cup I^k$. The variables of the problems are the following:

- z_{ik} $k \in K$, $i \in O^k$ a binary variable defining whether the shipment k is produced (or sourced) at node i;
- u_{ijk}, $(i,j) \in A$, $k \in K$ binary variable equal to 1 if shipment k is routed through arc (i, j), 0 otherwise;
- x_{ijk}, $(i,j) \in A$, $k \in K$ the quantity of shipment k routed through arc (i, j); and
- w_{ik}, $i \in N$, $k \in K$ the arrival time at node i for shipment k.

The integrated model can be formulated as follows:

$$\min \sum_{k \in K}\sum_{(i,j) \in A} wsu_k c_{ij} x_{ijk} + \sum_{k \in K}\sum_{(i,j) \in A} fr_{ijk} u_{ijk} + \sum_{k \in K}\sum_{i \in O^k}\sum_{j \in \Delta^+(i)} p_{ik} x_{ijk} + \sum_{k \in K}\sum_{i \in O^k}\sum_{j \in \Delta^+(i)} f_i z_{ik} + \alpha \sum_{(i,j) \in A}\sum_{k \in K} wsu_k e_{ij} x_{ijk} \quad (1)$$

subject to

$$\sum_{i \in O(k)}\sum_{j \in \Delta^+(i)} x_{ijk} = \sum_{d \in D^k} q_d \quad \forall k \in \overline{K} \quad (2)$$

$$\sum_{i \in \Delta^-(d)} x_{idk} = q_d \quad \forall d \in D^k, \quad k \in \overline{K} \quad (3)$$

$$\sum_{i \in O^k}\sum_{j \in \Delta^+(i)} x_{ijk} - \sum_{j \in D^k}\sum_{i \in \Delta^-(j)} x_{ijk} = 0 \quad \forall k \in K \backslash \overline{K} \quad (4)$$

$$\sum_{j \in \Delta^-(i)} x_{jik} - \sum_{j \in \Delta^+(i)} x_{ijk} = 0, \quad \forall k \in K, \forall i \in I^k \quad (5)$$

$$\sum_{j \in \Delta^-(i)} x_{jih} - \sum_{(h,k) \in B}\sum_{j \in \Delta^+(i)} x_{ijk} = 0 \quad \forall i \in O, \quad h \in K \backslash \overline{K} \quad (5')$$

$$w_{ik} + s_i + t_{ijk} - w_{jk} \le (1 - u_{ijk})M \quad \forall k \in K, \quad \forall (i,j) \in A \quad (6)$$

$$w_{dh} + s_d + lts_k - w_{dk} \le 0 \quad \forall (h,k) \in B, \quad d \in D^h \quad (6')$$

$$a_i \le \sum_{k \in K}\sum_{j \in \Delta^+(i)} r_{ik} x_{ijk} \le b_i, \quad \forall i \in O \quad (7)$$

$$E \le w_{ik} \le L, \quad \forall k \in K, \quad \forall i \in \{O, D\} \quad (8)$$

$$x_{ijk} \le C_{ijk} u_{ijk} \quad \forall (i,j) \in A, \quad k \in K \quad (9)$$

$$\sum_{i \in O(k)}\sum_{j \in \Delta^+(i)} x_{ijk} \le z_{ik} PC_{ik} \forall i \in N, \quad \forall k \in K \quad (10)$$

$$x_{ijk} \ge 0,\ z_{ik} \in \{0,1\}, \quad u_{ijk} \in \{0,1\}, \quad \forall k \in K, \forall (i,j) \in A,\ w_{ik} \ge 0\ \forall i \in N, \quad k \in K \quad (11)$$

The objective function (1) consists of five terms. The first and the second term express the variable and the fixed transportation cost respectively. The third and fourth terms consider, respectively, production and specific product-plant activation costs, while the fifth term considers the emission factors. Constraints (2)–(5′)

Table 2
Production cost for products and components in each supply and assembly site.

Production cost							
(EUR/unit)	Prod. 1	Prod. 2	Prod. 3	Comp. 1	Comp. 2	Comp. 3	Comp. 4
k	*1*	*2*	*3*	*4*	*5*	*6*	*7*
Supplier 1				300			
Supplier 2				240			
Supplier 3					30		
Supplier 4					26		
Supplier 5						80	110
Supplier 6						100	121
Assembly plant 1	2000	3400	1200				
Assembly plant 2	1700	3570	1116				
Assembly plant 3	2200	3128	1380				

Prod.=product and Comp.=component.

Table 4
Transportation cost by mode per tonne-kilometre (tkm).
Source: Schade et al. (2006)

Transport cost (EUR/tkm) in EU-25	
Road mode (heavy duty vehicle)	0.14
Rail mode	0.11
Short-sea shipping mode	0.009

Table 3
Production capacity for products and components in each supply and assembly site.

Production capacity							
(Units/year)	Prod. 1	Prod. 2	Prod. 3	Comp. 1	Comp. 2	Comp. 3	Comp. 4
k	*1*	*2*	*3*	*4*	*5*	*6*	*7*
Supplier 1				360,000			
Supplier 2				400,000			
Supplier 3					800,000		
Supplier 4					900,000		
Supplier 5						700,000	450,000
Supplier 6						1,300,000	750,000
Assembly plant 1	160,000	100,000	250,000				
Assembly plant 2	150,000	187,500	600,000				
Assembly plant 3	150,000	480,000	2,400,000				

Prod.=product and Comp.=component.

Table 5
Carbon dioxide (CO_2) emissions per tonne-kilometre (tkm).
Source: French Republic – Ministry of Ecology, Sustainable Development and Energy, and Environment and Energy Management Agency (2012)

Emissions (gCO_2/tkm)	
Road (semi-trailer truck with a GCW of 40 tonnes, large volumes, road diesel)	93.1
Rail	17.4
Short-sea shipping (Ro-Ro)	101.0

Table 6
Weight of products and components.

	Prod. 1	Prod. 2	Prod. 3	Comp. 1	Comp. 2	Comp. 3	Comp. 4
k	*1*	*2*	*3*	*4*	*5*	*6*	*7*
Weight (tonnes)	1.25	1.7	0.8	0.25	0.05	0.1	0.2

Prod.=product and Comp.=component.

characterize the flow to be followed by shipment k. In particular, constraints (5′) states that a shipment h is activated only if the related shipment k is activated with $(h, k) \in B$. Constraints (6)–(8) guarantee lead time feasibility with respect to time considerations. In particular, the constraint (6′) imposes time synchronisation between shipments following the BOM definition; constraint (7) imposes bounds on the time of activation of nodes, and constraint (8) imposes bounds on the total lead time of the supply chain. Constraints (9) and (10) guarantee feasibility with respect to capacity considerations while constraints (11) define the binary decision's domain.

5. Test instances and experiments

The test campaign was carried out by envisioning a hypothetical manufacturing network of the automotive industry consisting of facilities located in Europe. More specifically, six suppliers, three assembly plants and seven demand areas are considered in the model. The demand areas widely cover the European continent. Assembly plants are located in France, Germany and Italy. The suppliers are located in Western, Central, Southern and Eastern Europe. The explicit localisation of hypothetical supply, demand and production sites supports the quantification of real distances and travel times between the different countries considered (Fig. 3).

According to the BOMs representing the assembly options for four sub-systems (i.e., the components) necessary for three final products (three types of vehicle), nine facilities in total are considered in the test instances. The automotive industry case is particularly relevant because the environmental impacts of distribution have been scarcely investigated (Nieuwenhuis et al., 2012) and the supply chains can be highly complex. Nieuwenhuis et al. (2012), for example, applied to a cost model the CO_2 emissions in relation to distances in Korea–US and Korea–Europe automotive supply chains.

The market demand is expressed in terms of units of products (Products 1, 2 or 3) requested by the customers located in seven geographical areas (i.e., the demand areas). The demand data for finished products are inspired by publicly available sales data of European automotive industry. Under the assumption to include the most relevant sub-systems of the first level of a car BOM (e.g., engine, transmission, body/chassis), each product needs one unit of each component included in the specific BOM (Table 1). Therefore, the demand for each component can be accordingly obtained.

Hypotheses about the production costs of products and components are formulated for each component and product (Table 2). These values vary per each component and supplier or assembler. They do not refer to any specific, real firm or industrial case. Accordingly, the production capacity at each supplier's or assembler's site is also envisioned (Table 3).

In the test instances, 87 modal transport connections for the single-modal scenario settings and 213 for the multimodal ones are considered. In the multimodal scenario settings three transport modes are introduced: rail, road and short-sea shipping. The transportation operations are assumed to be carried out by means of transportation services. A transportation service is a capacitated transportation operation linking an origin with a destination in a particular time window by using a specific transport mode.

The emission parameters and cost for each transport mode and vehicle type are extracted from French Republic – Ministry of Ecology, Sustainable Development and Energy, and Environment and Energy Management Agency (2012) and (Schade et al. (2006). Tables 4 and 5 include the selected values. The impact of emissions on costs is quantified by 15 USD/tonne CO_2 (see, e.g., Government of Canada, 2008) and converted to 11.24 EUR/tonne CO_2 in the baseline scenarios (exchange rate of August 20, 2014). Transport costs are average operating costs per unit of transport performance (tonne-km) for each mode. They consist of the following components: wear and tear, personnel, capital cost, energy cost, insurance, infrastructure charge, other taxes and, charges, additional cost (Schade et al., 2006).

Transportation costs vary according to the distance to be covered. The transportation cost for each mode is presented in Table 4. The distances and travel times from the origins to the destinations were obtained through the elaboration of the output of the Google Earth™ functionalities. In the model, additional service times represent transhipment between different transport modes or vehicles. They range from 0.06 to 7 h depending on the potential modal shift. The environmental impact of freight transport considers the CO_2 emissions that are expressed in gCO_2/tonne-km. The CO_2 emission for each mode is presented in Table 5. To each product and component a weight is associated in order to compute the transportation capacity utilisation as well as the emissions of each shipment (Table 6).

The capacity of the different transportation means is expressed as capacity of the transportation services that can be provided throughout 1-year time horizon. The lead time of the supply chain is constrained to be up to 45 days while lts_k is equal to 24 h.

The model has been also used to quantify the impact of links capacity and plants production capacity on transportation costs and emissions. To this aim, some tests considers restrictive link capacity constraints for the connection of the shipments related to product and components, while other tests consider limited production capacity on nodes. Moreover flow splits are allowed. Under these settings, the objective function can be used to evaluate investments in link capacity and in production capacity in order to reduce transportation costs and emissions. It must however be underlined that the objective function considers only the operating cost of transport and the CO_2 emission cost. Other external cost components of transport such as congestion, accidents and noise are not

considered. Regarding the solution approach to the presented optimisation problem, a solver tool has been used.

5.1. Test scenarios

Four scenario settings (A, B, C and D) are considered. The first scenario setting (scenario setting A) is based on a single-modal network except for the arcs that connect the assembly plants with the demand area 7 (short-sea shipping mode). Each scenario setting is tested varying the activation of each cost component (i.e., production, transportation and emission factor costs) through the test parameter β_i, $i=1, 2, 3$. $\beta_i=1$ if the cost component i ($i=1$ refers to the transportation cost; $i=2$ to the production cost; $i=3$ to the cost impact of CO_2 emissions) is not considered (activated) in the objective function, 0 otherwise. Furthermore, an additional fixed cost for the activation of each transportation service is introduced in part of the experiments. The production costs are fixed in the base scenario setting. In this setting, the available capacity of both transportation arcs and production resources is equal to 100%. Table 7 summarises the experiments for the first scenario setting.

The second scenario setting (scenario setting B) is based on a multimodal network that includes the three transport mode options, i.e., road, rail and sea. Furthermore, the capacity of transportation services for each arc as well as production plants is set to 100%. The other characteristics are the same of the scenario setting A. Table 8 summarises the experiments for the scenario setting B.

The third scenario setting (scenario setting C) is based on the capacitated multimodal network introduced in the setting B but more stringent capacity constraints are introduced for the

Table 7
Scenario setting A.

Scenario setting A	Network	Fixed activation cost for transport (EUR)	Transportation arc capacity (%)	Production capacity (%)	CO_2 trade value (EUR/tCO_2)	Activation of cost components in the objective function[a]
Scen. 1	Unim.	0	100	100	11.24	$\beta_1=1$; $\beta_2=1$; $\beta_3=1$
Scen. 1.1	Unim.	0	100	100	11.24	$\beta_1=1$; $\beta_2=1$; $\beta_3=0$
Scen. 1.2	Unim.	0	100	100	11.24	$\beta_1=0$; $\beta_2=1$; $\beta_3=1$
Scen. 1.3	Unim.	0	100	100	11.24	$\beta_1=0$; $\beta_2=0$; $\beta_3=1$
Scen. 2	Unim.	150	100	100	11.24	$\beta_1=1$; $\beta_2=1$; $\beta_3=1$
Scen. 2.1	Unim.	150	100	100	11.24	$\beta_1=1$; $\beta_2=1$; $\beta_3=0$
Scen. 2.2	Unim.	150	100	100	11.24	$\beta_1=0$; $\beta_2=1$; $\beta_3=1$
Scen. 2.3	Unim.	150	100	100	11.24	$\beta_1=0$; $\beta_2=0$; $\beta_3=1$

Unim.=unimodal and Scen.=scenario.

[a] $z_1=1$, transportation cost component activated, 0 otherwise; $z_2=1$, production cost component activated, 0 otherwise; $z_3=1$, cost of CO_2 emissions activated, 0 otherwise.

Table 8
Scenario setting B.

Scenario setting B	Network	Fixed activation cost for transport (EUR)	Transportation arc capacity (%)	Production capacity (%)	CO_2 trade value (EUR/tCO_2)	Activation of cost components in the objective function[a]
Scen. 3	Multim.	0	100	100	11.24	$\beta_1=1$; $\beta_2=1$; $\beta_3=1$
Scen. 3.1	Multim.	0	100	100	11.24	$\beta_1=1$; $\beta_2=1$; $\beta_3=0$
Scen. 3.2	Multim.	0	100	100	11.24	$\beta_1=0$; $\beta_2=1$; $\beta_3=1$
Scen. 3.3	Multim.	0	100	100	11.24	$\beta_1=0$; $\beta_2=0$; $\beta_3=1$
Scen. 4	Multim.	150	100	100	11.24	$\beta_1=1$; $\beta_2=1$; $\beta_3=1$
Scen. 4.1	Multim.	150	100	100	11.24	$\beta_1=1$; $\beta_2=1$; $\beta_3=0$
Scen. 4.2	Multim.	150	100	100	11.24	$\beta_1=0$; $\beta_2=1$; $\beta_3=1$
Scen. 4.3	Multim.	150	100	100	11.24	$\beta_1=0$; $\beta_2=0$; $\beta_3=1$

Multim.=multimodal and Scen.=scenario.

[a] $z_1=1$, transportation cost component activated, 0 otherwise; $z_2=1$, production cost component activated, 0 otherwise; $z_3=1$, cost of CO_2 emissions activated, 0 otherwise.

Table 9
Scenario setting C.

Scenario setting C	Network	Fixed activation cost for transport (EUR)	Transportation arc capacity (%)	Production capacity (%)	CO_2 trade value (EUR/tCO_2)	Activation of cost components in the objective function[a]
Scen. 5-base	Multim.	150	100	100	11.24	$\beta_1=1$; $\beta_2=1$; $\beta_3=1$
Scen. 5.1	Multim.	150	90	100	11.24	$\beta_1=1$; $\beta_2=1$; $\beta_3=1$
Scen. 5.2	Multim.	150	80	100	11.24	$\beta_1=1$; $\beta_2=1$; $\beta_3=1$
Scen. 5.3	Multim.	150	70	100	11.24	$\beta_1=1$; $\beta_2=1$; $\beta_3=1$
Scen. 5.4	Multim.	150	60	100	11.24	$\beta_1=1$; $\beta_2=1$; $\beta_3=1$

Multim.=multimodal and Scen.=scenario.

[a] $z_1=1$, transportation cost component activated, 0 otherwise; $z_2=1$, production cost component activated, 0 otherwise; $z_3=1$, cost of CO_2 emissions activated, 0 otherwise.

Table 10
Scenario setting D.

Scenario setting D	Network	Fixed activation cost for transport (EUR)	Transportation arc capacity (%)	Production capacity (%)	CO_2 trade value (EUR/tCO_2)	Activation of cost components in the objective function[a]
Scen. 6-base	Multim.	150	100	100	11.24	$\beta_1=1$; $\beta_2=1$; $\beta_3=1$
Scen. 6.1	Multim.	150	100	94	11.24	$\beta_1=1$; $\beta_2=1$; $\beta_3=1$
Scen. 6.2	Multim.	150	100	87	11.24	$\beta_1=1$; $\beta_2=1$; $\beta_3=1$
Scen. 6.3	Multim.	150	100	80	11.24	$\beta_1=1$; $\beta_2=1$; $\beta_3=1$

Multim.=multimodal and Scen.=scenario.

[a] $z_1=1$, transportation cost component activated, 0 otherwise; $z_2=1$, production cost component activated, 0 otherwise; $z_3=1$, cost of CO_2 emissions activated, 0 otherwise.

Table 11
Results of the scenario setting A.

Scenario setting A	Transport cost (mEUR)	Production cost (mEUR)	CO_2 emissions cost (mEUR)	Total cost (mEUR)
Scenario 1	158.57	1410.34	1.44	1570.35
Scenario 1-1	158.57	1410.34	1.44	1568.91
Scenario 1-2	165.03	1408.42	1.51	1409.93
Scenario 1-3	110.18	1580.31	1.10	1.10
Scenario 2	158.57	1410.34	1.44	1570.36
Scenario 2-1	158.57	1410.34	1.44	1568.92
Scenario 2-2	165.05	1408.42	1.51	1409.93
Scenario 2-3	110.19	1580.31	1.10	1.10

Table 12
Results of the scenario setting B.

Scenario setting B	Transport cost (mEUR)	Production cost (mEUR)	CO_2 emissions cost (mEUR)	Total cost (mEUR)
Scenario 3	143.70	1412.02	1.27	1556.99
Scenario 3-1	143.70	1412.02	1.28	1555.72
Scenario 3-2	175.24	1408.42	1.20	1409.62
Scenario 3-3	150.65	1559.24	0.68	0.68
Scenario 4	143.71	1412.02	1.27	1557.00
Scenario 4-1	143.71	1412.02	1.27	1555.73
Scenario 4-2	175.27	1408.42	1.20	1409.62
Scenario 4-3	150.68	1559.24	0.68	0.68

transportation arcs. The purpose is to observe the effects of the reduction in the capacity of transportation arcs. The fixed activation cost for transport services is included in all the scenario experiments. However, all the cost components of the objective function are activated in this setting, i.e., $z_i=1$, $i=1, 2, 3$. The scenario characteristics related to the available production capacity (equal to 100%) and the CO_2 emissions are the same of the scenario settings A and B.

The base scenario (scenario 5-base) is equal to the scenario 4 in the setting B in order to evaluate the difference from a baseline in terms of effects of transportation capacity variation. Table 9 summarises the experiments for the scenario setting C.

The fourth scenario setting (scenario setting D, in Table 10) includes a variation in the production capacity while the capacity of the transportation arcs is equal to 100%. The purpose is to analyse the effects of the reduction in the capacity of production resources. The other characteristics are the same of the scenario setting C. Note that the reduction in the production capacity is limited to 20% because for a larger reduction the problem solutions become infeasible.

The base scenario (scenario 6-base) is equal to the Scenario 4 and 5-base, respectively, in the settings B and C. So doing, the effects of production capacity variation from (i) the baseline Scenario 4 as well as (ii) the experiments of the scenario setting C can be assessed. In Table 10 the experiments for the scenario setting D are presented.

Table 13
Results of the scenario setting C.

Scenario setting C	Transport cost (mEUR)	Production cost (mEUR)	CO_2 emissions cost (mEUR)	Total cost (mEUR)
Scenario 5-base	143.71	1412.02	1.27	1557.00
Scenario 5-1	145.05	1412.02	1.28	1558.35
Scenario 5-2	146.56	1412.02	1.29	1559.87
Scenario 5-3	148.07	1412.02	1.30	1561.40
Scenario 5-4	148.55	1416.27	1.31	1566.13

Table 14
Results of the scenario setting D.

Scenario setting D	Transport cost (mEUR)	Production cost (mEUR)	CO_2 emissions cost (mEUR)	Total cost (mEUR)
Scenario 6-base	143.71	1412.02	1.27	1557.00
Scenario 6-1	143.20	1416.16	1.26	1560.62
Scenario 6-2	141.88	1431.05	1.25	1574.18
Scenario 6-3	137.64	1440.56	1.22	1579.43

Table 15
Computational times.

Test	Execution time (s)
$N=50$; $V=250$. PN=17	4.6
$N=50$; $V=250$. PN=22	4.88
$N=50$; $V=250$. PN=27	15.37
$N=50$; $V=250$. PN=32	41.25
$N=50$; $V=250$. PN=37	46.21
$N=50$; $V=250$. PN=42	200.00[a]

PN=production node options.

[a] No optimal solution found in time limit of 200 s (gap=33%).

6. Computational results

The model has been implemented on IBM ILOG CPLEX Optimization Studio®. The case study presented in this article consists of a production network of 16 nodes in the single-modal setting and 44 in the multimodal one; the arcs considered are 87 in the single-modal

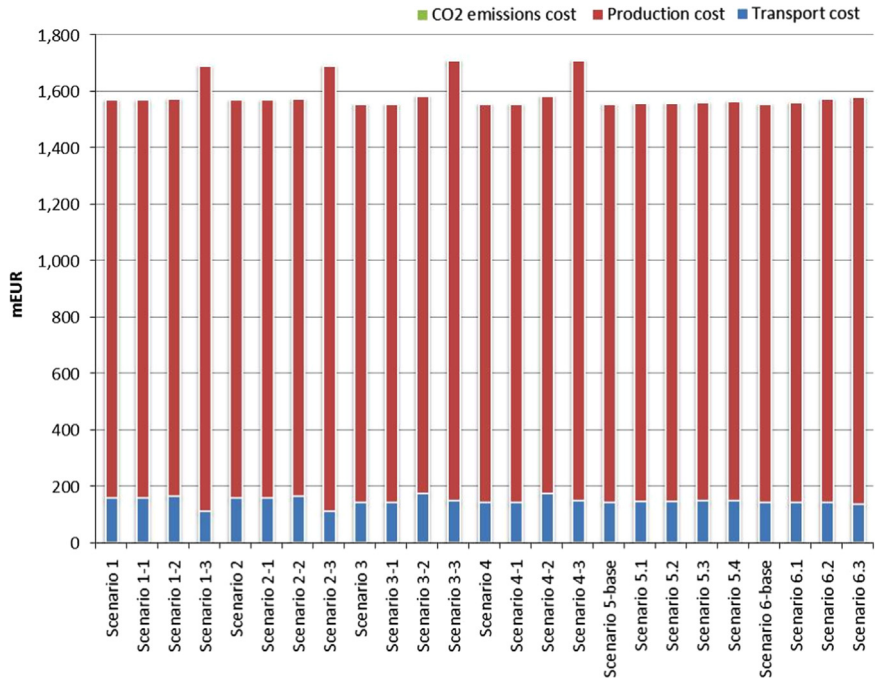


Fig. 4. Total costs in all the scenario settings.

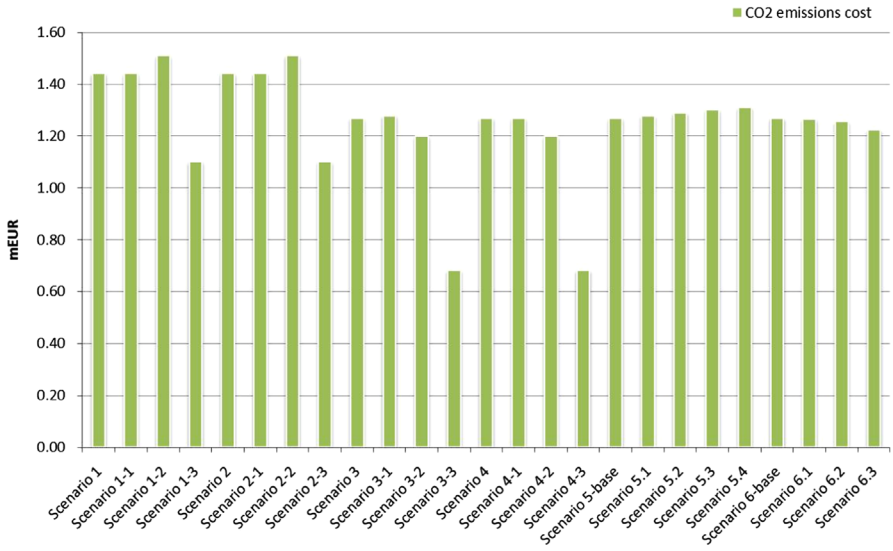


Fig. 5. CO_2 emission costs in all the scenario settings.

Table 16
Effect of multimodality in scenario settings A and B.

	Min cost[a]	Max cost[a]	Gap cost	Min emission[b]	Max emission[b]	Gap emission
Unimodal	1570.35	1691.61	7.7%	1.10	1.51	37.1%
Multimodal	1556.99	1710.59	9.9%	0.68	1.28	87.7%
Gap multimodal–unimodal	−0.9%	1.1%		−38.1%	−15.3%	

[a] mEUR.
[b] mEUR Trade cost for CO_2 with trade cost equal to 11.24 EUR/tCO_2.

settings and 213 in the multimodal ones; The shipments consist of three finished products and four subassemblies; specific constraints state alternative sources for finished products and components. The test instances were solved in less than a second on a Windows PC with Intel® Core™ i7-2670QM 2.20 GHz processor with 8 GB RAM. The results are presented in Table 11 for scenario A, in Table 12 for scenario B, in Table 13 for scenario C and in Table 14 for scenario D.

The small computational time is due to the particular topology of the network with restricted routing options and due to the high-level aggregation of the BOM entailing a reduced number of components and related sourcing/production options. Pre-configured source-destination paths and clustering of markets in conjunction with BOM aggregation can help solve the model that is indeed not polynomial. To clarify this point, further tests were

carried out by increasing the number of available nodes (PN) where the production of finished products as well as subassemblies can take place according to the original BOM configuration.

A network consisting of $N=50$ nodes with a mean node degree equal to five and $V=250$ arcs was randomly generated. The results highlighted an exponential increase in the computational time as presented in Table 15. A further experiment was carried out over a network consisting of 100 nodes and 500 arcs while not achieving a solution in a reasonable time.

7. Discussion

7.1. The effect of multimodality on costs and emissions

Figs. 4 and 5 present the total costs and the CO_2 emission costs for all the scenario settings. By comparing the scenario settings A and B, which respectively include single-modal and multimodal transport, it can be observed that multimodal arcs could benefit the distribution network in terms of both economic advantages and emissions. The model has been used to quantify this benefit. Results are then grouped in Table 16 by considering, for each scenario, the sum of all cost categories independently of the value of β_i. The comparisons among the single-modal (unimodal) and multimodal scenario settings using the full transportation and production capacities are highlighted by the negative gap values in the third row of the table. The results put into evidence the high, positive impact of multimodality on the emissions performance. In fact, when the objective function minimises only the emission component (Scenarios 3-3 and 4-3) the multimodal network can improve the emission performance by 38.1% with respect to the unimodal one that is based on road transport. However, considering the total cost including all the cost components (the "min cost" column) the improvement is equal only to 0.9%.

More specifically, it should be also noticed that in the scenarios where only the CO_2 cost emission component is activated (i.e., Scenarios 1-3, 2-3, 3-3, and 4-3), a relevant decrease also in the transportation cost (Fig. 6) is obtained together with the correspondent reduction in CO_2 emissions and related costs (Fig. 7). The total costs however increase in these scenarios due to the production cost component (Fig. 4).

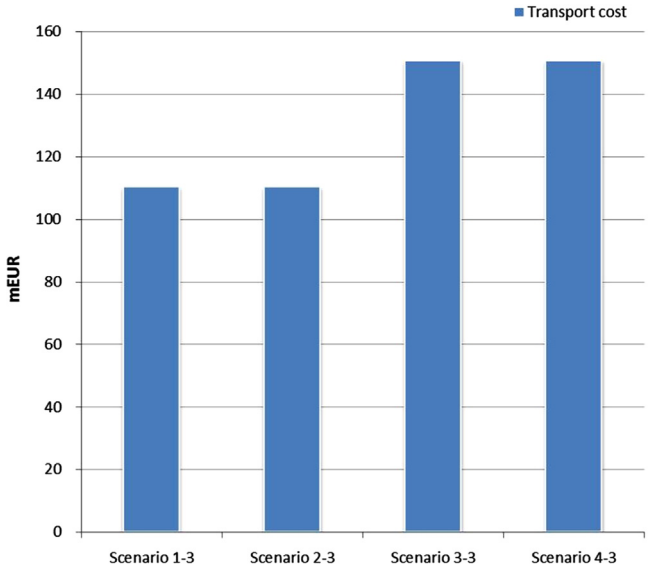


Fig. 6. Transportation costs in the scenario settings with only the CO_2 cost component activated in the objective function.

7.2. The effect of arc and production capacity on costs and emission

In Figs. 8–10 the transportation and emission costs when all the cost components are activated in the objective function are presented. Fig. 8 highlights the different scenarios when the transportation capacity varies while Fig. 9 includes also the scenarios when the production capacity as well as the transportation capacity decrease. Fig. 10 presents the more detailed CO_2 cost trends all these scenario settings. Accordingly, the production cost trends are presented in Fig. 11, where production costs are indicated in the scenario settings with all the cost components activated in the objective function and the transportation capacity as well as the production capacity vary (Scenario settings C and D, respectively). The reduction in arc capacity implies a high impact on the cost of CO_2 emission which rises 3.3% when capacity is equally reduced on arcs by 40%, while transportation costs are 3% higher (Scenario setting C). The results are totally different when the production capacity of plants is reduced (Scenario setting D). In this case, if the reduction is equal to 20%, the production costs rise only 2% and the transportation costs are lower by 4.2% (CO_2

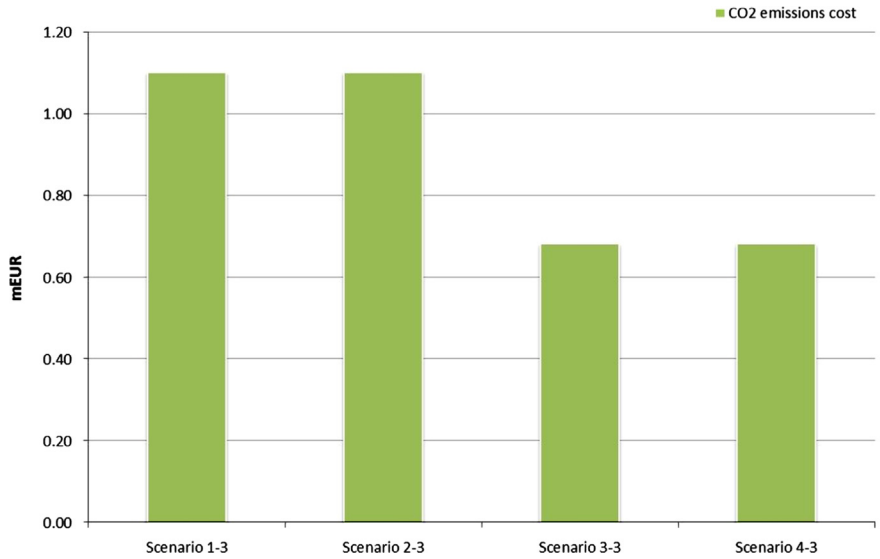


Fig. 7. CO_2 emission costs in the scenario settings with only the CO_2 cost component activated in the objective function.

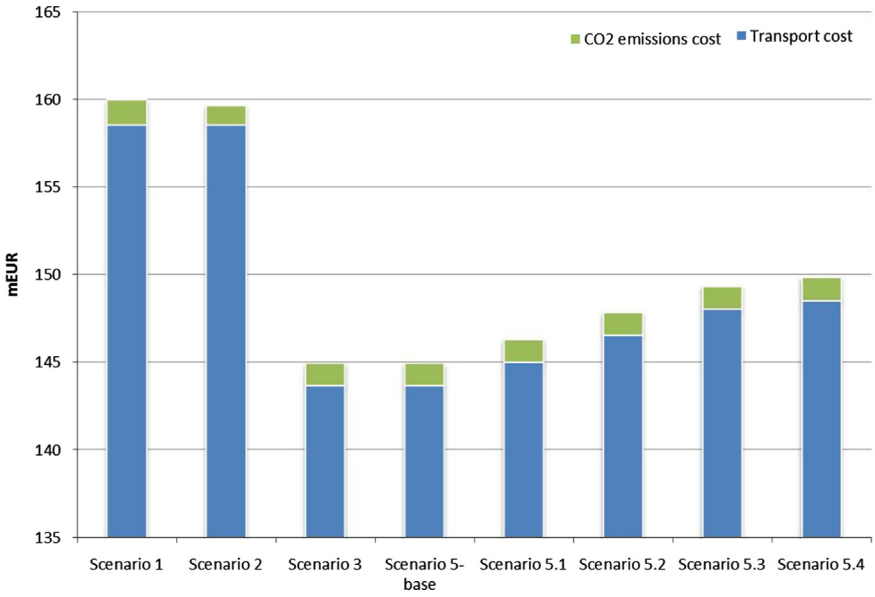


Fig. 8. Transportation and CO_2 emission costs with all the cost components activated.

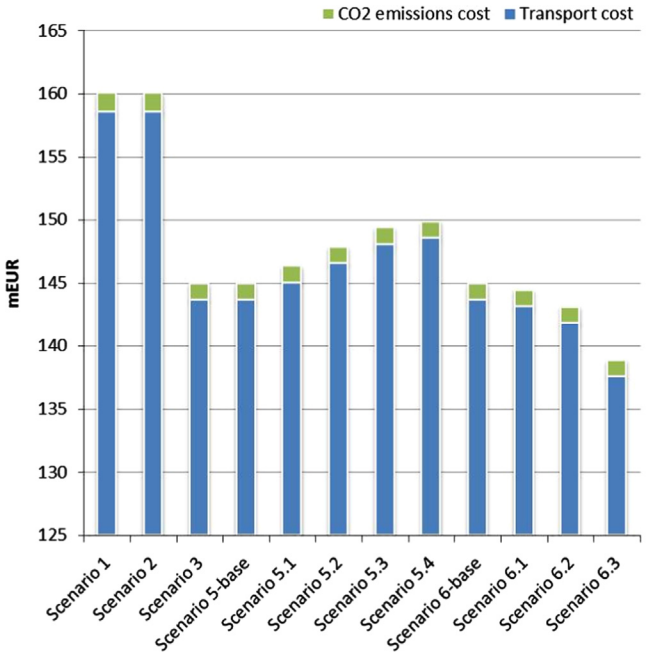


Fig. 9. Transportation and CO_2 emission costs with transportation and production capacity variation.

emission costs lower by 3.6%). The insight is that the 2% of production costs represents approximately 16% of the transportation costs. Indeed, a further experiment has been conducted to analyse the performance of the system when only the transportation cost is activated. The results have been compared to the Scenario 4.3, where only the emission costs are considered. In this experiment it can be argued that the emission gain is remarkable (−70%). However, the transport cost component accounts for a reduction by approximately 35%. It can be then demonstrated that the use of multimodality leverages the emission reduction.

The model can be used also to evaluate the relocation options of industrial facilities and the re-design of the supply network. The cost parameters affecting the production costs can lead to completely different network configurations. In this particular case, the high production costs do not leave significant, relative space for obtaining very high savings in terms of both CO_2 and transportation costs. However, in this work transportation costs are operating cost to which only the calculation of the CO_2 emission component has been added to the objective function. The additional, external costs of transport for society and environment are due to other pollutants, congestion, accidents and noise. They are not considered in the present study. Their quantification requires specific elaborations but their inclusion would definitely strengthen the analysis of freight transport in sustainable supply chains.

Furthermore, it would be extremely interesting to analyse cases where the logistics and transportation costs cover a higher share of the total product cost. This is particularly important in the case of production offshoring. Nevertheless, it should be considered that when the relocation involves developing countries with low labour cost, additional cost items should be estimated and included in the related parameters because of increased coordination costs, risks or potential hidden, unexpected cost items. Additionally, the offshoring decision would also imply increase in CO_2 emissions, transportation costs, inventory costs, and reduced flexibility due to longer lead times to serve the demand areas of the presented scenario settings. In our vision, the model is aimed at fostering production models regionally-based and backsourcing strategies for reindustrialisation in order to simultaneously address the economic and environmental dimensions of sustainable manufacturing. Furthermore, the model allows estimating the impacts on network design and management decisions deriving from economic incentives related to CO_2 emissions for freight transport. In particular we found that the considered trade value for a tonne of CO_2 (equal to EUR 11.24) seems to be too low in the freight transportation sector. A different incentive scheme should be adopted in order to substantially augment environmentally friendlier transportation.

8. Conclusions and future research

In this article a model integrating production networks and sustainable freight transportation for strategic and tactical decisions is presented. The optimisation model includes BOM constraints with alternative supply sources having different production and distribution costs. The objective function concurrently

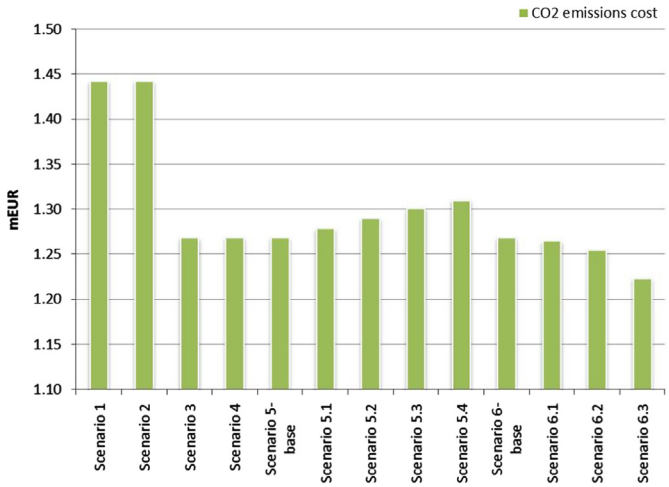


Fig. 10. Detail of the CO_2 emission costs with transportation and production capacity variation.

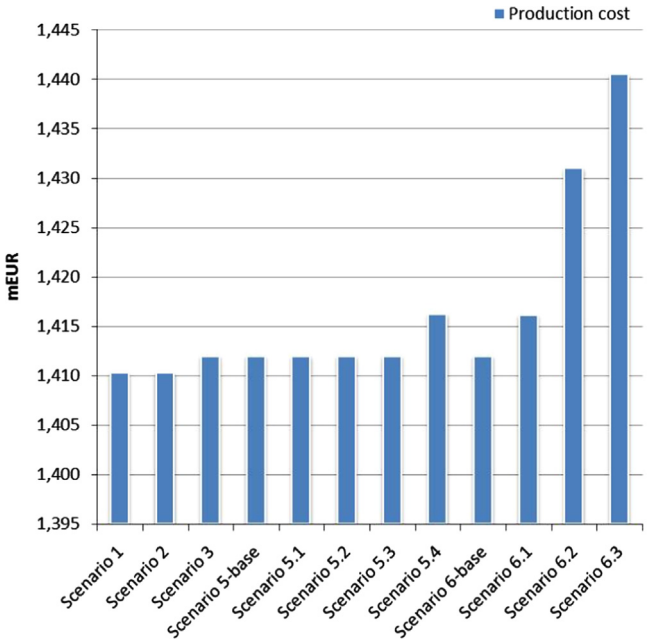


Fig. 11. Production costs in the scenario settings with transportation and production capacity variation.

considers supply, production and transportation costs as well as the environmental impact of transport over a multimodal network. The model was developed in order to act as a tool to study the effect of production relocation and transport multimodality on the economic and environmental performance of the system. To test the usefulness of the model we used a set of test cases envisioning a scenario of the automotive manufacturing and distribution industry in Europe. In particular the production and distribution of a three product-based automobile production have been studied. It has been demonstrated that the introduction of more multimodal distribution options allows pursuing environmental and cost reduction objectives. Multimodality can therefore leverage reductions in CO_2 emissions and become a source of competitive advantage. A work limitation is the underlying complexity of the developed MILP model. The execution was very fast on the test case instances. Nevertheless, a higher computational time is needed to run larger scale instances representing real data made of several finished products, with possibly hundreds of components and several sourcing options. Moreover, it will be interesting to study real test instances where transportation operators may offer several multimodal distribution options. Future research will consider data aggregation strategies and extension of the model to consider multi-period scenarios and production emissions. Social impacts will also be investigated.

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